TOOL-AUGMENTED LLM DIALOGUE OPTIMIZATION

DiaTool-DPO: Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

Model TA-LLM interaction as a 5-state MDP, auto-construct chosen/rejected multi-turn trajectories, and improve slot-filling + tool rejection without human preference labeling.

Audience: ML Researchers | Audience: NLP Engineers | Theme: RLHF Alignment

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

OBSERVED FAILURE MODES

- Fails to ask follow-up questions when mandatory slots are missing.
- Hallucinates tool invocations despite incomplete arguments.
- Fails to reject out-of-scope queries, causing unsafe or irrelevant calls.

WHY SFT IS INSUFFICIENT

- SFT imitates local responses but lacks trajectory-level preference control.
- No explicit optimization for call/ask/reject decisions across dialogue states.
- Production TA-LLMs need policy-level alignment over multi-turn paths.